

Algorithms and Data Structures

Algorithms Used

Content Filtering Algorithm

Input: Content approval status

Output: Only approved content displayed

Logic:

```
if content.status == "approved":  
    display  
else:  
    hide
```

Search Algorithm

Keyword-based filtering using database queries

Optimized using indexed fields

Activity Logging Algorithm

Triggered on content access

Stores timestamp, user ID, and content ID

Data Structures

Relational database tables:

ChildProfile (user data)

Content (educational materials)

ActivityLog (tracking usage)

Lists/QuerySets used for retrieving filtered content